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Student Number

2017

Mathematics Ext 2

HSC Course Assessment Task 3

May 25 2017

General Instructions

- Reading time – 5 minutes
- Working time 1 hour
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- In Questions 6 to 8 show relevant mathematical reasoning and/or calculations
- Answer each question in a separate writing booklet
- This paper must not be removed from the examination room
- A reference sheet is provided at the back of this paper
- Diagrams are NOT to scale

Total Marks – 40

Section I - Pages 2 - 4

5 marks

- Attempt Questions 1 to 5
- Allow about 8 minutes for this section

Section II - Pages 5 to 7

35 marks

- Attempt Questions 6 to 8
- Allow about 52 minutes for this section

Multiple choice	/5
Q6	/16
Q7	/10
Q8	/9
Total	/40

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 20% of the HSC Course Assessment

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Section I

5 marks

Attempt Questions 1 to 4

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for questions 1 to 4 (Detach from paper).

1. The equation $x^3 - y^3 + 3xy + 1 = 0$ defines y implicitly as a function of x . What is the value of $\frac{dy}{dx}$ at the point $(-1, 3)$?
- (A) $\frac{1}{2}$
- (B) $\frac{2}{5}$
- (C) $\frac{2}{3}$
- (D) $\frac{1}{5}$
2. A directrix of an ellipse has the equation $x = \frac{25}{4}$ and one of its foci has the coordinate $(-4, 0)$. What is the equation of this ellipse?
- (A) $\frac{x^2}{5} + \frac{y^2}{3} = 1$
- (B) $\frac{x^2}{3} + \frac{y^2}{5} = 1$
- (C) $\frac{x^2}{25} + \frac{y^2}{9} = 1$
- (D) $\frac{x^2}{9} + \frac{y^2}{25} = 1$
3. $S(4, 0)$ is a focus of the rectangular hyperbola $x^2 - y^2 = k$. Which of the following is the value of k ?
- (A) $2\sqrt{2}$
- (B) $4\sqrt{2}$
- (C) 8
- (D) 32

4. If the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are inclined to each other at an angle α , then $\tan \alpha$ is equal to:

(A) $\frac{2a}{|a^2 - b^2|}$

(B) $-\left|\frac{2ab}{a^2 - b^2}\right|$

(C) $\frac{2b}{|a^2 - b^2|}$

(D) $\frac{2ab}{|a^2 - b^2|}$

5. Which integral is equal to

$$\int_{-a}^a f(x) dx$$

(A) $\int_0^a f(x) - f(-x) dx$

(B) $\int_0^a f(x) - f(a - x) dx$

(C) $\int_0^a f(x - a) + f(-x) dx$

(D) $\int_0^a f(x - a) + f(a - x) dx$

Section II

35 marks

Attempt Questions 6 to 8

Allow about 52 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 6 to 8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (16 marks) Use a SEPARATE writing booklet

- (a) Find 1

$$\int \frac{1}{x(\ln x)^2} dx$$

- (b) By completing the square, find 2

$$\int \frac{1}{x^2 + 6x + 10} dx$$

- (c) (i) Find real numbers a, b and c such that 2

$$\frac{3x+1}{(x-1)(x^2+1)} = \frac{a}{x-1} + \frac{bx+c}{x^2+1}$$

- (ii) Hence, find $\int \frac{3x+1}{(x-1)(x^2+1)} dx$ 2

- (c) Evaluate $\int_1^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx$ 3

- (d) Find $\int \cos^5 x \, dx$ 3

- (e) Using the substitution $t = \tan \frac{x}{2}$, find $\int \frac{1}{1 + \sin x - \cos x} dx$ 3

End of Question 6

Question 7 (10 marks) Use a SEPARATE writing booklet

- (a) With respect to the Cartesian axes, the line $x = -1$ is a directrix, and the point $(-2, 0)$ is a focus with eccentricity $\sqrt{2}$.

i) Derive from first principles the equation of the conic. 2

ii) Given that the equation of the tangent to the curve at any point $P(x_1, y_1)$ on it is $xx_1 - yy_1 = 2$. 2

If the tangent cuts the x -axis at $(X, 0)$ and the y -axis at $(0, Y)$, prove that the locus of the point $R(X, Y)$ is $2Y^2 - 2X^2 = X^2Y^2$

(b)

Consider the hyperbolas $H_1 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $H_2 : \frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$.

i) Derive the equations of the asymptotes (standard formulae are not to be used). 2
What do you notice?

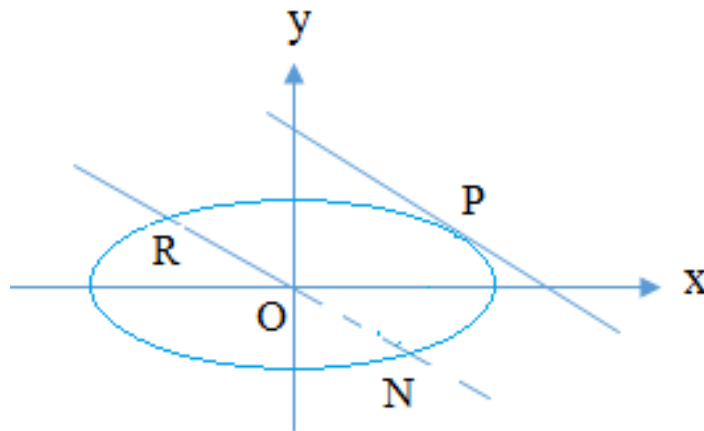
ii) $P(a \sec \theta, b \tan \theta)$ is a point on H_1 . Derive the equation of the tangent at P . 2

iii) The tangent at P to H_1 intersects H_2 at Q and R . The tangents at Q and R to H_2 intersect at T . 2
Show that T lies on H_1 .

End of Question 7

Question 8 (9 marks) Use a SEPARATE writing booklet

- (a) $P(a\cos\theta, b\sin\theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. A line through the origin and parallel to the tangent at P, meets the ellipse at R.



- i) Find the equation of OR. 2
- ii) Prove that the area of $\triangle OPR$ is $\frac{1}{2}ab$ for all position of P. 3

- (b) Let

$$I_n = \int_0^1 x^n \sqrt{1-x^3} \, dx, \text{ for } n \geq 2.$$

- (i) Show that $I_n = \frac{2n-4}{2n+5} \times I_{n-3}$, for $n \geq 5$. 3
- (ii) Hence find I_8 1

End of Examination

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Student Number



Preliminary / HSC Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 10. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↗

- Start Here** →
1. A ☐ B ☐ C ☐ D ☐
 2. A ☐ B ☐ C ☐ D ☐
 3. A ☐ B ☐ C ☐ D ☐
 4. A ☐ B ☐ C ☐ D ☐
 5. A ☐ B ☐ C ☐ D ☐

Q1 B $3x^2 - 3y^2 \frac{dy}{dx} + 3y + 3x \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{3x^2 + 3y}{3y^2 - 3x}$ $= \frac{3 + 9}{27 + 3} = \frac{2}{5}$ Q2 C $\frac{a}{e} = \frac{25}{4}, ae = 4 \rightarrow a = 5 \text{ and } e = \frac{4}{5}$ $(A) b^2 = 25(1 - (\frac{4}{5})^2) \rightarrow b = 3$ Equation is $\frac{x^2}{25} + \frac{y^2}{9} = 1$	Q4 D Equations of asymptotes $y = \pm \frac{b}{a}x$ $\tan \alpha = \left \frac{\frac{b}{a} - \frac{b}{a}}{1 + \frac{b}{a} \times -\frac{b}{a}} \right $ $= \left \frac{\frac{2b}{a}}{1 - \frac{b^2}{a^2}} \right = \left \frac{2ab}{a^2 - b^2} \right $ $= \frac{2ab}{ a^2 - b^2 } \quad a \& b > 0$
Q3 C $e = \sqrt{2} \quad S(\sqrt{2}k, 0) \quad \therefore \quad \sqrt{2}k = 4 \quad \rightarrow \quad k = 8$ $a = \sqrt{k}$	Q5 D $\int_{-a}^a f(x) dx = \int_0^a f(x) dx + \int_{-a}^0 f(x) dx$ $\int_0^a f(x - a) dx \quad \text{let } u = x - a \text{ then } du = dx$ $= \int_{-a}^0 f(u) du \quad x = 0, u = -a \text{ and } x = -a, u = 0$ $= \int_{-a}^0 f(x) dx$ $\int_0^a f(a - x) dx \quad \text{let } u = a - x \text{ then } du = -dx$ $= - \int_a^0 f(u) du \quad x = 0, u = a \text{ and } x = a, u = 0$ $= \int_0^a f(x) dx$ <i>removed</i>

$$\text{6a. } \int \frac{1}{x(\ln x)^2} dx$$

$$\text{let } u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\therefore \int \frac{1}{x} \cdot \frac{1}{(\ln x)^2} dx$$

$$= \int \frac{du}{u^2} \cdot \frac{1}{u^2} dx$$

$$= \int \frac{du}{u^2} \cdot \frac{1}{u^2} \cdot \frac{dx}{u^2}$$

$$= \int u^{-2} du$$

$$= \frac{u^{-1}}{-1} = -u^{-1} = \frac{-1}{u}$$

$$= \frac{-1}{\ln x}$$

very well done
although some made the
error

$$\int \frac{1}{u^2} du = \frac{1}{u^3} + c$$

X

wrong!

$$\frac{b}{\int \frac{1}{x^2+6x+10} dx}$$

$$= \int \frac{1}{(x^2+6x+9)+1} dx$$

$$= \int \frac{1}{(x+3)^2+1} dx \quad [1 \text{ mark}]$$

$$= \tan^{-1}(x+3) + C \quad [1 \text{ mark}]$$

very well done

$$\frac{c}{(x-1)(x^2+1)} = \frac{a}{x-1} + \frac{bx+c}{x^2+1}$$

with RHS = $\frac{a(x^2+1) + (x-1)(bx+c)}{(x-1)(x^2+1)}$

$$\therefore 3x+1 = ax^2 + a + bx^2 - bx + cx - c$$

$$ax^2 + bx^2 = 0x^2 \quad (\text{equating coefficients of } x^2)$$

$$\therefore a+b=0 \quad (1)$$

$$-bx + cx = 3x \quad (\text{equating coefficients of } x)$$

$$\therefore -b+c=3 \quad (2)$$

$$a-c=1 \quad (3) \quad (\text{equating constants})$$

from (1) $a=-b$

sub into (2) $-b+c=3 \Rightarrow a+c=3 \quad (4)$

$$(3) + (4) \quad a+c=3+$$

$$a-c=1$$

$$2a=4 \Rightarrow a=2$$

very well done
although some
minor algebraic errors

From ① $a+b=0 \Rightarrow 2+b=0 \Rightarrow \underline{b=-2}$

From ③ $a-c=1 \Rightarrow 2-c=1 \Rightarrow -c=-1 \therefore \underline{c=1}$

$$\therefore \frac{3x+1}{(x-1)(x^2+1)} = \frac{2}{x-1} + \frac{-2x+1}{x^2+1}$$

[2 marks a,b,c]

[1 mark 1 wrong]

$$\therefore \int \frac{3x+1}{(x-1)(x^2+1)} dx = \int \frac{2}{x-1} + \frac{-2x+1}{x^2+1} dx$$

$$= \int \frac{2}{x-1} dx + \int \frac{-2x}{x^2+1} dx + \int \frac{1}{x^2+1} dx$$

$$= 2 \ln(x-1) - \ln(x^2+1) + \tan^{-1}x + C$$

[2 marks

all correct]

[1 mark for

each error]

Well done although
some sign issues

d Evaluate $\int_1^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx$

let $x = 2 \sin \theta$

$\frac{dx}{d\theta} = 2 \cos \theta$

terminals

$\sqrt{3} = 2 \sin \theta$

$\sin \theta = \frac{\sqrt{3}}{2}$

$\sin \theta = \frac{1}{2}$

$\therefore \theta = \frac{\pi}{3}$

$\therefore \theta = \frac{\pi}{6}$

$\therefore \int_1^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{(2 \sin \theta)^2}{\sqrt{4-(2 \sin \theta)^2}} \cdot 2 \cos \theta d\theta$ [1 mark]

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{4 \sin^2 \theta}{\sqrt{4-4 \sin^2 \theta}} \cdot 2 \cos \theta d\theta$

using $\sqrt{1-\sin^2 \theta} = |\cos \theta|$

$= |\cos \theta|$

$= \cos \theta$ in 1st quad

$= \frac{1}{3} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{4 \sin^2 \theta}{2 \cos \theta} d\theta$

using

$\cos 2\theta = 1-2 \sin^2 \theta$

$2 \sin^2 \theta = 1-\cos 2\theta$

$= 2 \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1-\cos 2\theta d\theta$ [1 mark]

Many skipped steps
and lost signs / constants

$$= 2 \left[\frac{\theta - 2 \sin 2\theta}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= \left[2\theta - 2 \sin 2\theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \left(\frac{2\pi}{3} - \sqrt{3} \right) - \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right) = \frac{\pi}{3} \quad [1 \text{ mark}]$$

$$\textcircled{2} \int \cos^5 x \, dx$$

$$= \int \cos x \cdot \cos^4 x \, dx$$

$$= \int \cos x \cdot \cos^2 x \cdot \cos^2 x \, dx$$

using $\cos^2 x = 1 - \sin^2 x$

$$= \int \cos x \cdot (1 - \sin^2 x)(1 - \sin^2 x) \, dx$$

$$= \int \cos x (1 - 2\sin^2 x + \sin^4 x) \, dx \quad [1 \text{ mark}]$$

$$= \int \cos x - 2\sin^2 x \cdot \cos x + \sin^4 x \cos x \, dx$$

$$= \int \cos x \, dx - \int 2\sin^2 x \cos x \, dx + \int \sin^4 x \cos x \, dx \quad [1 \text{ mark}]$$

$$= \sin x - \left(\frac{2}{3} \sin^3 x \right) + \left(\frac{1}{5} \sin^5 x \right) + C$$

$\frac{d}{dx} \sin x = \cos x$ $\frac{d}{dx} \sin^3 x = 3 \sin^2 x \cos x$ $\frac{d}{dx} \sin^5 x = 5 \sin^4 x \cos x$

$$= \sin x - \frac{2}{3} \sin^3 x + \frac{1}{5} \sin^5 x + C \quad [1 \text{ mark}]$$

$$= \sin x - \frac{2 \sin^3 x}{3} + \frac{\sin^5 x}{5} + C \quad [1 \text{ mark}]$$

Generally well done
 → standard question
 learn the process

$$t = \tan\left(\frac{x}{2}\right)$$

$$\int \frac{1}{1 + \sin x - \cos x} dx$$

$$\sin x = \frac{2t}{1+t^2}$$

$$= \int \frac{1}{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \cdot \left(\frac{2}{1+t^2}\right) dt \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$= \int \frac{1}{1+t^2 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$t = \tan\left(\frac{x}{2}\right)$$

$$\tan^{-1} t = \tan^{-1}\left(\tan\left(\frac{x}{2}\right)\right)$$

$$\tan^{-1} t = \frac{x}{2}$$

$$\therefore x = 2 \tan^{-1} t$$

$$\frac{dx}{dt} = 2 \cdot \frac{1}{1+t^2}$$

$$\therefore dx = \frac{2}{1+t^2} dt$$

$$= \int \frac{1}{\frac{2t+2t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

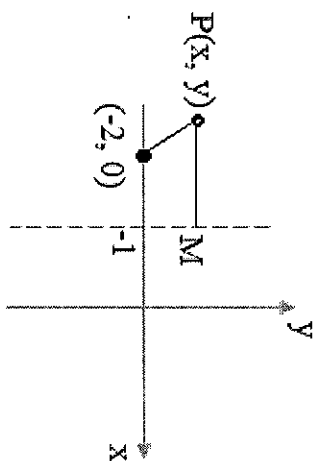
$$= \int \frac{(1+t^2)}{2t(1+t)} \cdot \frac{2}{(1+t^2)} dt$$

Generally very well done but those who skipped steps often made mistakes

$$\begin{aligned}
 &= \int \frac{1}{t(1+t)} \cdot dt \quad [1 \text{ mark}] \Rightarrow \left| \frac{1}{t(1+t)} = \frac{a}{t} + \frac{b}{1+t} \right. \\
 &= \int \frac{1}{t} dt + \int \frac{-1}{1+t} dt \quad [1 \text{ mark}] \quad \left. \begin{array}{l} a(1+t) + bt = 1 \\ a + at + bt = 1 \\ a = 1 \\ at + bt = 0t \end{array} \right\} \\
 &= \ln\left[\tan\left(\frac{x}{2}\right)\right] - \ln\left[1 + \tan\left(\frac{x}{2}\right)\right] + C \quad \left. \begin{array}{l} \therefore a+b=0 \\ \therefore b=-1 \end{array} \right\} [1 \text{ mark}]
 \end{aligned}$$

Question 7

(a)
(i)



$$\frac{PS}{PM} = \sqrt{2}$$

$$PS^2 = 2PM^2$$

$$\begin{aligned} (x+2)^2 + y^2 &= 2(-1-x)^2 \\ x^2 + 4x + 4 + y^2 &= 2x^2 + 4x + 2 \\ x^2 - y^2 &= 2 \end{aligned}$$

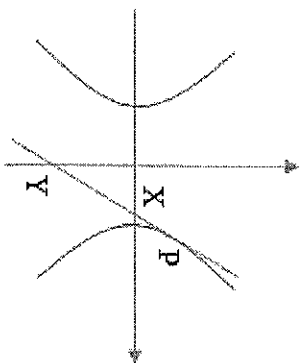
is the rectangular hyperbola.

1-relationship between PS & PM

1 - correct equation

Many students used the formula for $S(a, 0)$ and $x = \frac{a}{2}$ to arrive at the equation.

(ii)



$$\text{Sub } (x, 0) \text{ into } xx_1 - yy_1 = 2$$

$$x_1 = \frac{2}{x}$$

(1)

$$\text{Sub } (0, y) \text{ into } xx_1 - yy_1 = 2$$

$$y_1 = \frac{2}{y}$$

(2)

Since (x_1, y_1) lies on $x^2 - y^2 = 2$

Sub (1) & (2) into the equation

$$\Rightarrow \frac{4}{x^2} - \frac{4}{y^2} = 2$$

$$4x^2 - 4y^2 = 2x^2y^2$$

$$\therefore 2x^2 - 2y^2 = x^2y^2$$

1 - coordinates of x_1 and y_1 .

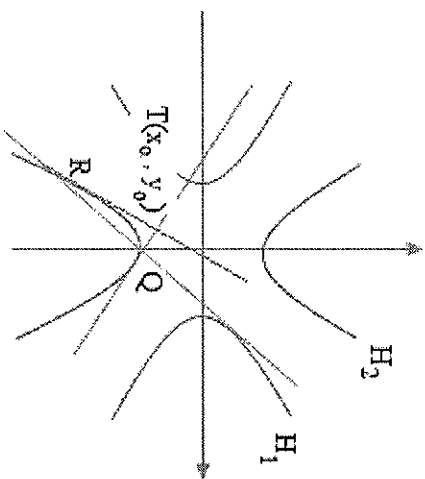
(student must derive the result from 1st principle to obtain the mark)

Well done!

1 - correct locus equation.

(b) (i) $\frac{y^2}{x^2} = 1$ for the asymptotes of H_1 $\Rightarrow x^2 y^2 - a^2 y^2 = a^2 b^2$ ($\div x^2$) $\Rightarrow b^2 - \frac{a^2 y^2}{x^2} = \frac{a^2 b^2}{x^2}$ $x \rightarrow \infty, \quad \frac{a^2 b^2}{x^2} \rightarrow 0$ $\therefore \frac{a^2 y^2}{x^2} = b^2 \Rightarrow y = \pm \frac{b}{a} x$ As Similarly the asymptotes of H_2 are $y = \pm \frac{b}{a} x$ Both H_1 & H_2 have the same asymptotes. (ii) $\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{b^2 x}{a^2 y}$ At ($a \sec \theta, b \tan \theta$) $m = \frac{b \sec \theta}{a \tan \theta}$ Equation of the tangent to H_1 is $\frac{y}{b} \tan \theta = \frac{x}{a} \sec \theta$ (1) $\Rightarrow \frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = \sec^2 \theta - \tan^2 \theta = 1$	1 - dividing by x^2 2 - correct equation of asymptotes For both H_1 & H_2 with conclusion	<i>You could also do</i> $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow \left \frac{x}{a} + \frac{y}{b} \right \left \frac{x}{a} - \frac{y}{b} \right = 1$ $\left \frac{b}{a} x + y \right = \frac{1}{\left \frac{x}{a} \pm \frac{y}{b} \right }$ $x \rightarrow \infty, y \rightarrow \infty, \pm \frac{b}{a} x \rightarrow y$ $y \rightarrow \pm \frac{b}{a} x$ <i>Most students get 1 mark here for making the correct conclusion.</i>
Well done.	1 - finding the gradient 1 - correct equation of tangent	

(iii)



Not drawn to scale

The equation of the tangent to H_1 , $\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$, is also the equation of the chord of contact RQ from $T(x_0, y_0)$.

The equation of the chord of contact from the point $T(x_0, y_0)$ is $\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$ (2)

$$\frac{\tan \theta}{b} = \frac{y_0}{b^2} \Rightarrow y_0 = b \tan \theta$$

$\therefore (x_0, y_0)$ lies on H_1 .

1 - explanation

1 - showing (x_0, y_0) lies on H_1

Except one student
no one got this right.
It was important to
know right from the
beginning RQ is a
chord of contact.

Question 8

$$I_n = \int_0^1 x^n \sqrt{1-x^3} dx$$

$$= \frac{1}{3} \int_0^1 x^{n-2} \cdot 3x^2 \sqrt{1-x^3} dx$$

$$\text{using } \int u \cdot dv = u \cdot v - \int v \cdot du$$

$$= \left[-\frac{1}{3} x^{n-2} \sqrt{(1-x^3)^3} \right]_0^1 + \frac{1}{3} x^{\frac{2}{3}} (n-2) \int_0^1 x^{n-3} (1-x^3) \sqrt{1-x^3} dx$$

[1 mark]

$$I_n = \frac{2(n-2)}{9} \int_0^1 x^{n-3} \sqrt{1-x^3} dx - \frac{2(n-2)}{9} \int_0^1 x^n \sqrt{1-x^3} dx$$

[1 mark]

$$I_n \left(1 + \frac{2(n-2)}{9} \right) = \frac{2(n-2)}{9} \cdot I_{n-3}$$

$$I_n \left(\frac{2n+5}{9} \right) = \left(\frac{2n-4}{9} \right) \cdot I_{n-3}$$

$$\therefore I_n = \frac{2n-4}{2n+5} \cdot I_{n-3} \quad [1 \text{ mark}]$$

2 marks : All correct.

1 mark: Applying integration by parts and significant progress to arrive at the reduction formula.

i) $I_8 = \frac{2 \times 8 - 4}{2 \times 8 + 5} \times I_5$

$$= \frac{12}{21} \times \frac{2 \times 5 - 4}{2 \times 5 + 5} \cdot I_2$$

$$= \frac{12}{21} \times \frac{6}{15} \cdot I_2$$

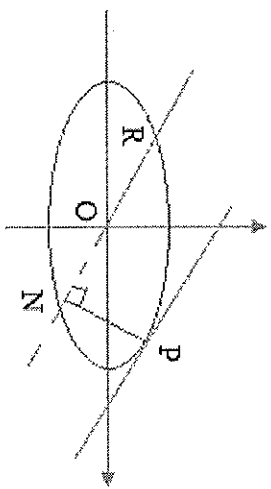
$$I_2 = \int_0^1 x^2 \sqrt{1-x^3} dx \quad \text{let } u = 1-x^3$$

$$= \frac{-1}{3} \int_0^1 -3x^2 \sqrt{1-x^3} dx \quad \frac{du}{dx} = -3x^2$$

terminals $x=0 \Rightarrow u=1$
 $x=1 \Rightarrow u=0$

$$= \frac{-1}{3} \int_1^0 u^{\frac{1}{2}} du \quad = \frac{-1}{3} \left[\frac{2u^{\frac{3}{2}}}{3} \right]_1^0 = \frac{-1}{3} \left(0 - \frac{2}{3} \right) = \frac{2}{9} \Rightarrow \therefore I_8 = \frac{12}{21} \times \frac{6}{15} \times \frac{2}{9} = \frac{16}{315}$$

(b)



$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$m = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta} = -\frac{b \cos \theta}{a \sin \theta}$$

At $(a \cos \theta, b \sin \theta)$

1 mark: Correct answer
from correct working.

This question was very well done.

Equation of OR is $y = -\frac{b \cos \theta}{a \sin \theta} x$ (1)

(ii)

Sub (1) into $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{a^2} + \frac{b^2 \cos^2 \theta}{b^2 a^2 \sin^2 \theta} = 1 \Rightarrow x = \pm a \sin \theta$

sub into (1) $\Rightarrow y = \pm b \cos \theta$

$OR = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$

Perpendicular distance PN to the line $\frac{b \cos \theta}{a \sin \theta} x + y = 0$ is :

$$\left| \frac{\frac{ab \cos^2 \theta}{a \sin \theta} + b \sin \theta}{\sqrt{\frac{b^2 \cos^2 \theta}{a^2 \sin^2 \theta} + 1}} \right| = \left| \frac{ab \cos^2 \theta + ab \sin^2 \theta}{a \sin \theta \sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right| = \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

Area $\Delta OPR = \frac{1}{2} \times \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}} \times \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$

$= \frac{1}{2} ab$

Which is independent of P (for all position of P).

1 mark was given for getting these points

For trying to find the perpendicular distance

1 mark was awarded
1 - correct area of triangle OPR
(Significant figures to in simplifying the expression).

1 - correct perpendicular distance

1 - correct equation of OR

1 - correct distance of OR

1 - correct gradient of OR

